

DSA0603 Data Handling and Visualization Slot B

CO2 – Assessment Test 6 (AT3)

Case Study

Instructions:

For each question, write the program in the specified language (R or Python), generate the required visualization, and answer the interpretation sub-questions clearly with justification.

Case Study 1 – Retail Sales Performance Dashboard Scenario

A retail company has collected one year of sales data from five regions (North, South, East, West, Central). The dataset contains monthly sales, product categories, customer segments, and total revenue. The management wants to identify:

- The highest-performing region.
- Monthly sales trends.
- Revenue contribution of each product category.
- Relationship between advertising expenditure and sales.
- Sales distribution across regions.

Tasks

1. Select the most suitable visualization for each requirement:
 - Regional revenue comparison
 - Monthly sales trend
 - Product category contribution
 - Advertising expenditure vs sales
 - Regional sales distribution
2. Using **R**, create:
 - Bar Plot
 - Stacked Bar Chart
 - Histogram

- Density Plot
- Scatter Plot

3. Interpret:

- Which region generated the highest revenue?
- Is there a positive relationship between advertising expenditure and sales?
- Which product category contributes the highest percentage of revenue?

Dataset:

```
sales <- data.frame(
  Region=c("North","South","East","West","Central"),
  Revenue=c(450000,380000,420000,500000,390000),
  Advertising=c(45,38,40,55,36),
  Electronics=c(120000,90000,100000,140000,95000),
  Furniture=c(100000,85000,90000,120000,80000),
  Clothing=c(80000,70000,75000,90000,70000),
  Grocery=c(150000,135000,155000,150000,145000)
)
```

```
monthly_sales <- data.frame(
  Month=month.abb,
  Sales=c(32000,35000,38000,41000,45000,47000,
  49000,52000,50000,53000,55000,60000)
)
```

1. Suitable Visualizations

Requirement	Visualization
Regional revenue comparison	Bar Plot
Monthly sales trend	Line Chart

Product category contribution

Stacked Bar Chart

Advertising vs Sales

Scatter Plot

Regional sales distribution

Histogram

2. R Programs

Bar Plot

```
barplot(sales$Revenue,  
names.arg=sales$Region,  
col="skyblue",  
main="Regional Revenue")
```

Stacked Bar Chart

```
category<-as.matrix(sales[,4:7])  
  
barplot(t(category),  
col=c("red","blue","green","orange"),  
legend=rownames(t(category)),  
main="Product Category Contribution")
```

Histogram

```
hist(sales$Revenue,  
col="lightgreen",  
main="Revenue Distribution")
```

Density Plot

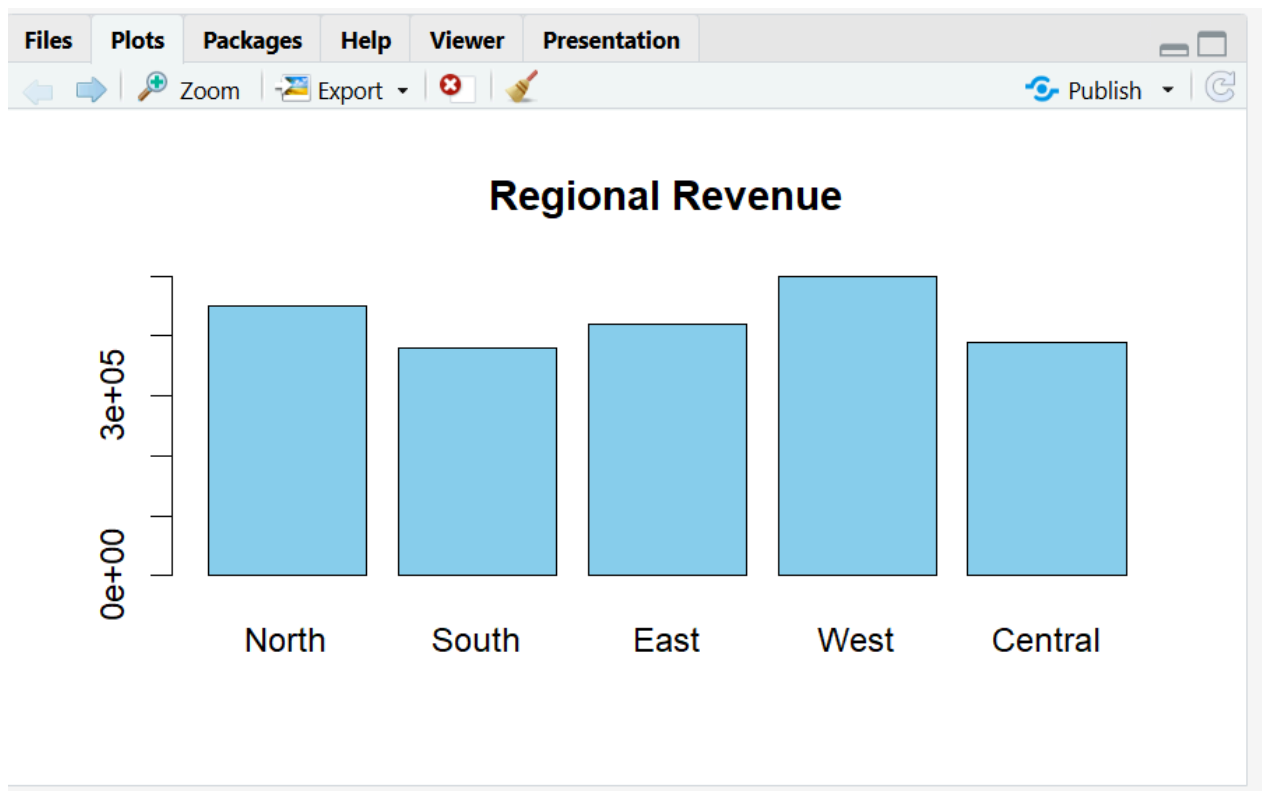
```
plot(density(sales$Revenue),  
main="Revenue Density")
```

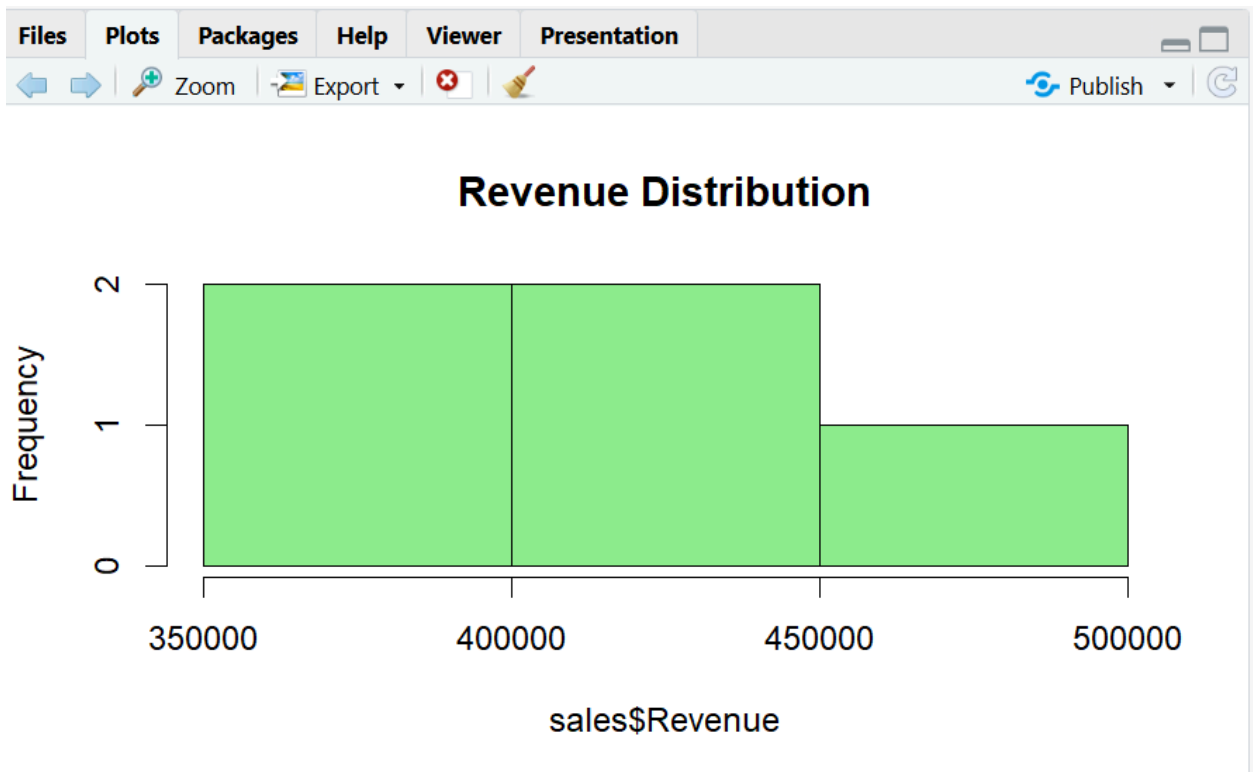
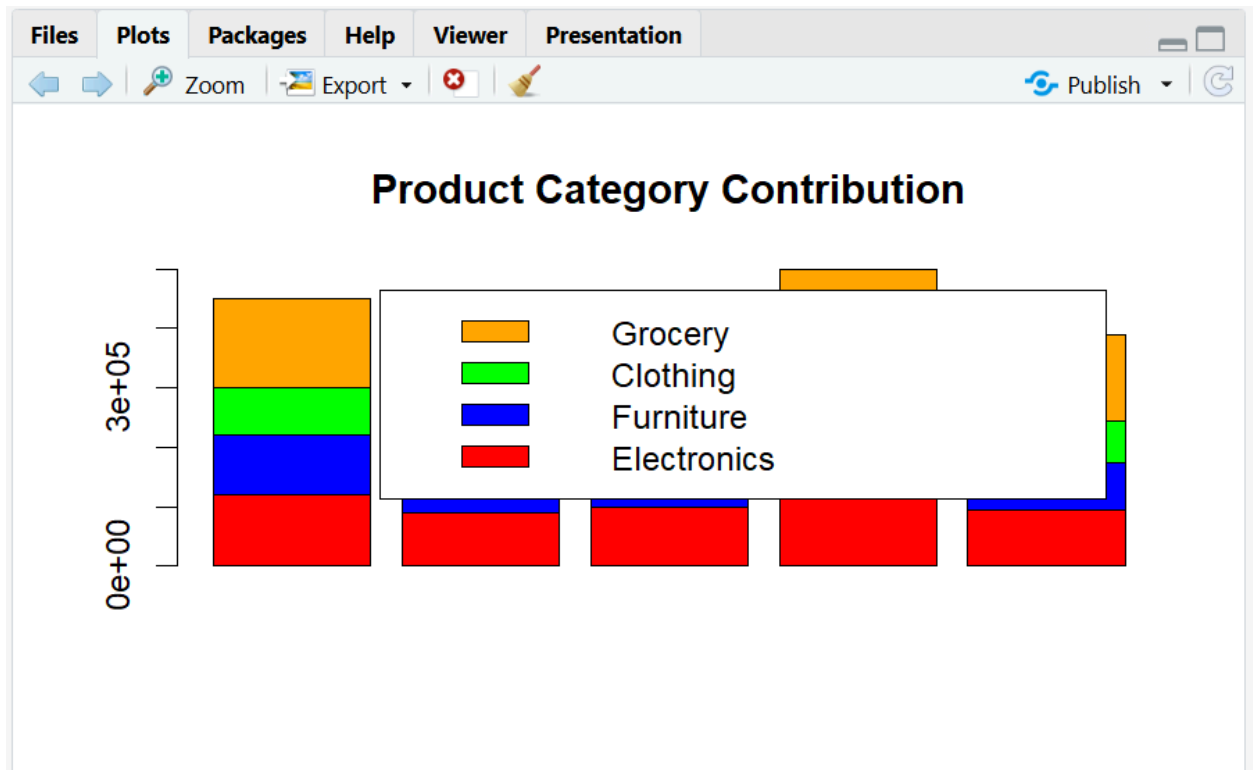
Scatter Plot

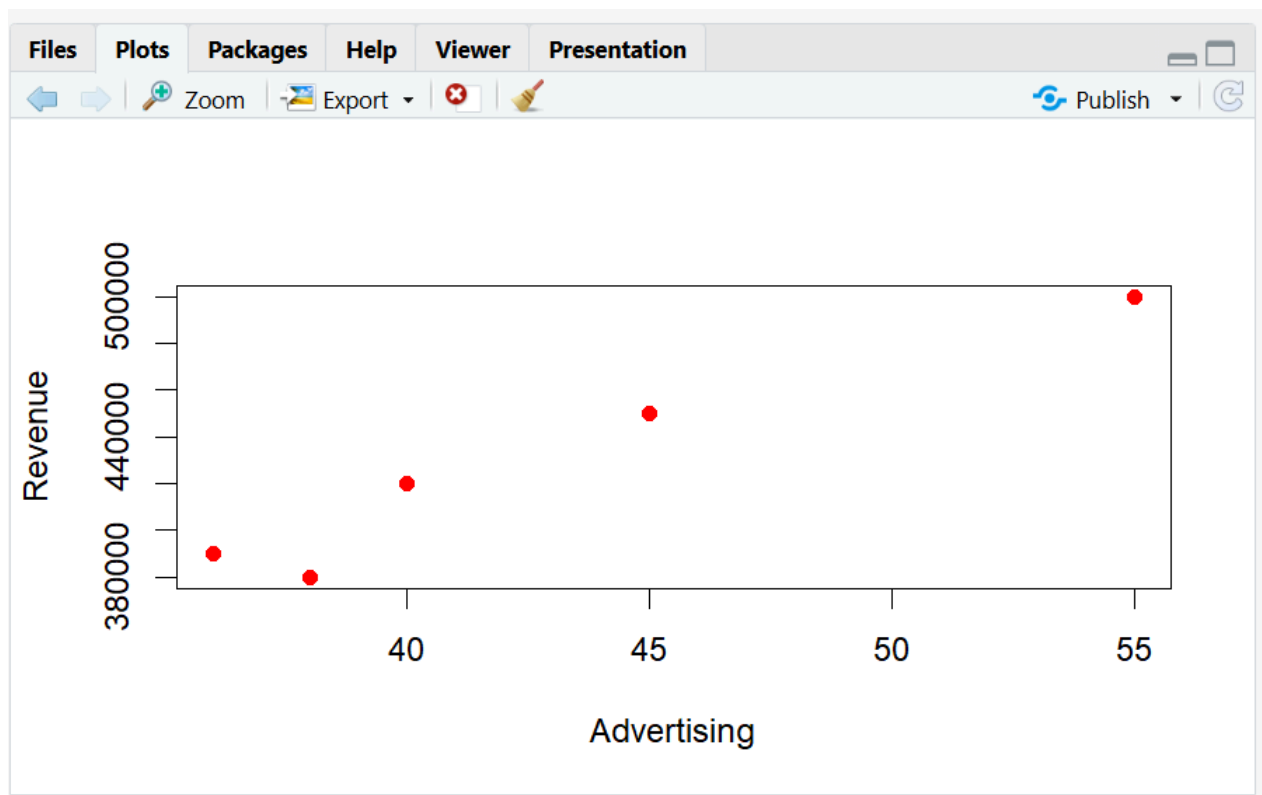
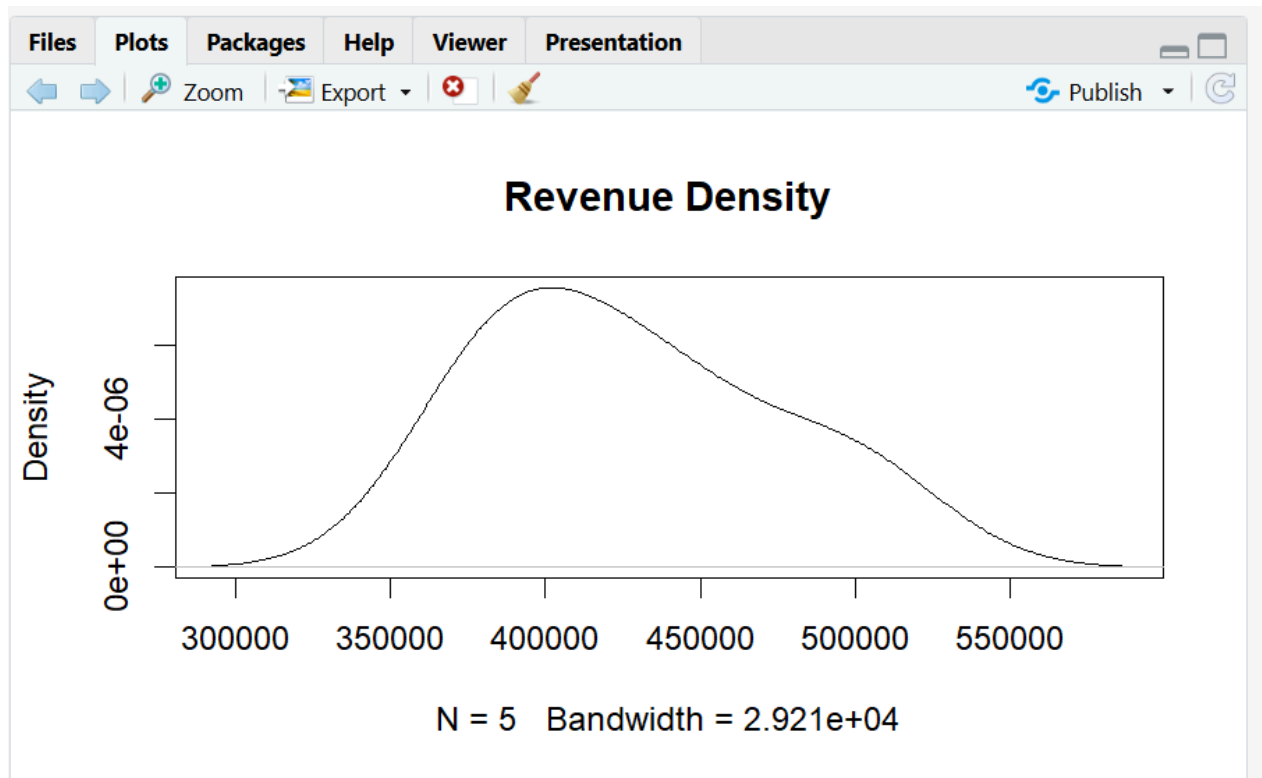
```
plot(sales$Advertising,  
sales$Revenue,  
pch=19,  
col="red",  
xlab="Advertising",  
ylab="Revenue")
```

3. Interpretation

- Highest revenue: **West Region (₹500,000)**.
- Advertising and sales show a **positive relationship**.
- **Electronics** contributes the highest percentage of revenue.







Case Study 2 – University Student Performance Analysis

Scenario

A university stores examination data for four departments: • Computer

Science

- Information Technology
- Electronics
- Mechanical

The dataset includes:

- Semester marks
- Attendance percentage
- CGPA
- Placement status

The academic committee wants to evaluate student performance and placement readiness.

Tasks

1. Choose appropriate visualizations for:
 - Department-wise average marks
 - Distribution of CGPA
 - Placement percentage
 - Attendance vs CGPA relationship
 - Semester-wise performance trend
2. Develop the following in **R**:
 - Grouped Bar Chart
 - Violin Plot
 - Histogram with Density Curve
 - Scatter Plot
 - Line Chart
3. Interpret:
 - Which department has the highest average CGPA? ○ Does higher

attendance improve CGPA?

- Which department has the highest placement rate?

Dataset

```
students<-data.frame(  
  Department=c("CS","IT","ECE","MECH"),  
  Marks=c(88,82,80,75),  
  Attendance=c(92,90,87,85),  
  CGPA=c(8.9,8.6,8.2,7.8),  
  Placement=c(95,90,85,80)  
)
```

```
semester<-data.frame(  
  Semester=1:8,  
  AverageMarks=c(72,74,76,79,81,83,85,88)  
)
```

Suitable Visualizations

Requirement	Visualization
Average Marks	Grouped Bar Chart
CGPA Distribution	Histogram
Placement %	Violin Plot
Attendance vs CGPA	Scatter Plot
Semester Performance	Line Chart

R Programs

Grouped Bar Chart

```
barplot(rbind(students$Marks,students$Placement),  
beside=TRUE,  
names.arg=students$Department,  
col=c("blue","green"))
```

Violin Plot

```
library(ggplot2)  
  
ggplot(students,  
aes(x=Department,  
y=Placement))+  
geom_violin(fill="orange")
```

Histogram with Density

```
hist(students$CGPA,  
probability=TRUE,  
col="lightblue")  
  
lines(density(students$CGPA),  
col="red",lwd=2)
```

Scatter Plot

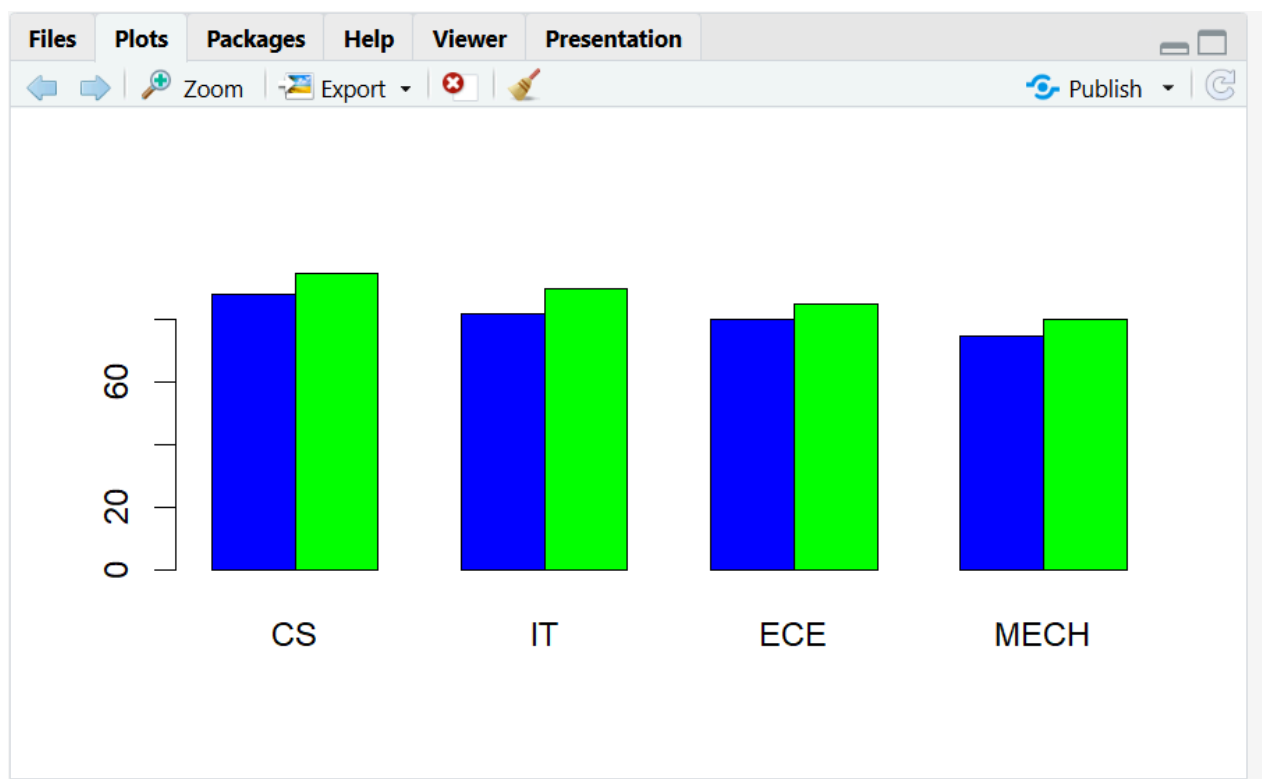
```
plot(students$Attendance,  
students$CGPA,  
pch=19,  
col="darkgreen")
```

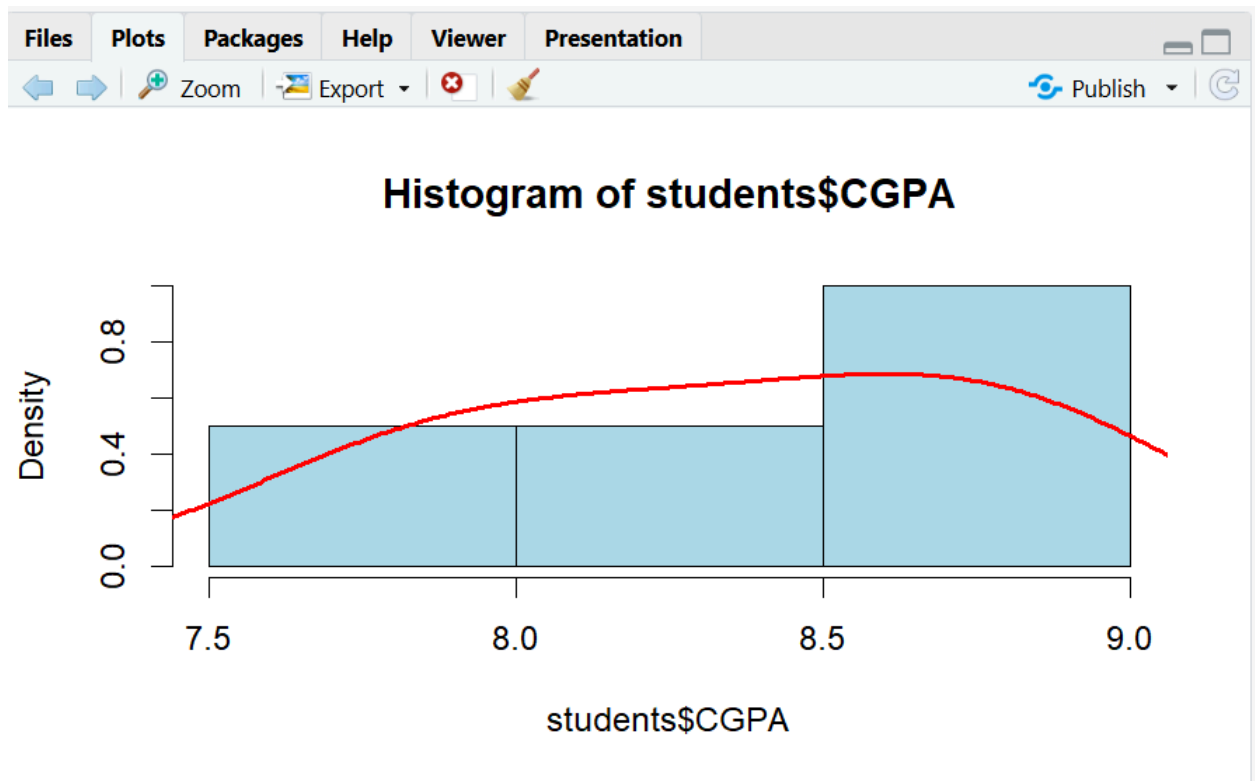
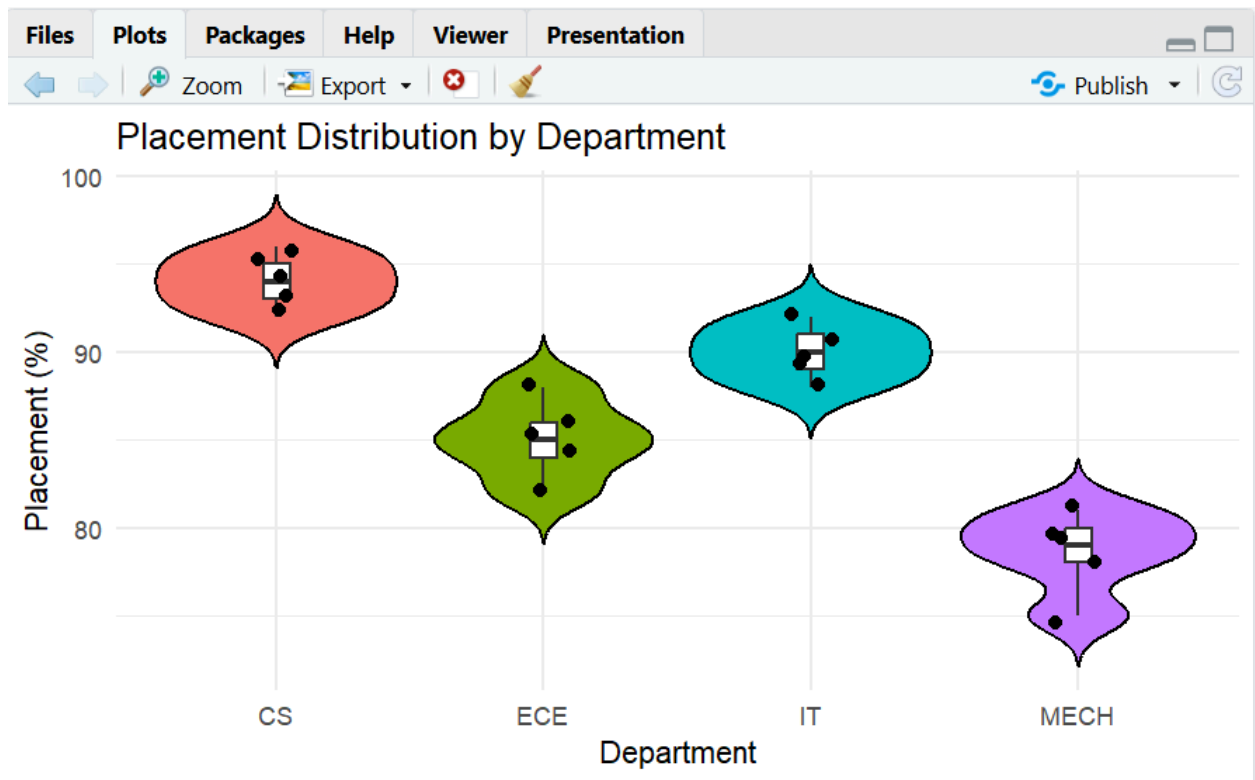
Line Chart

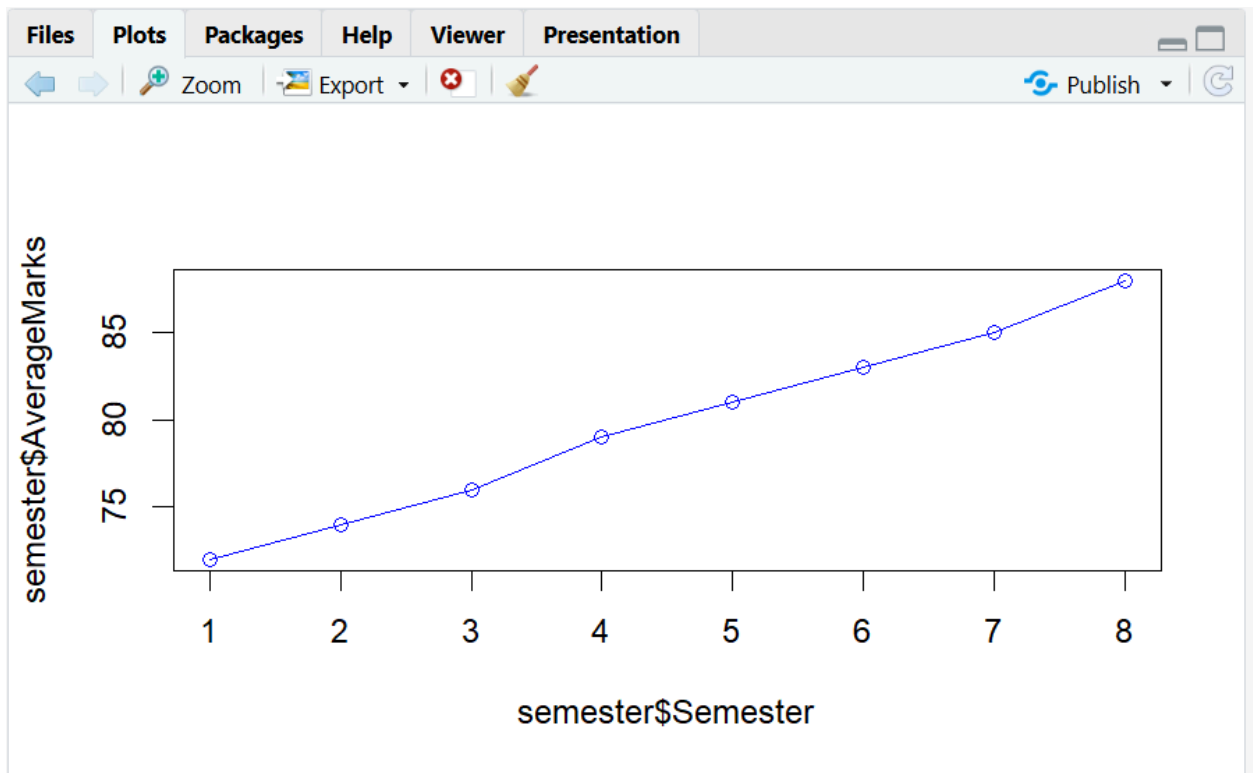
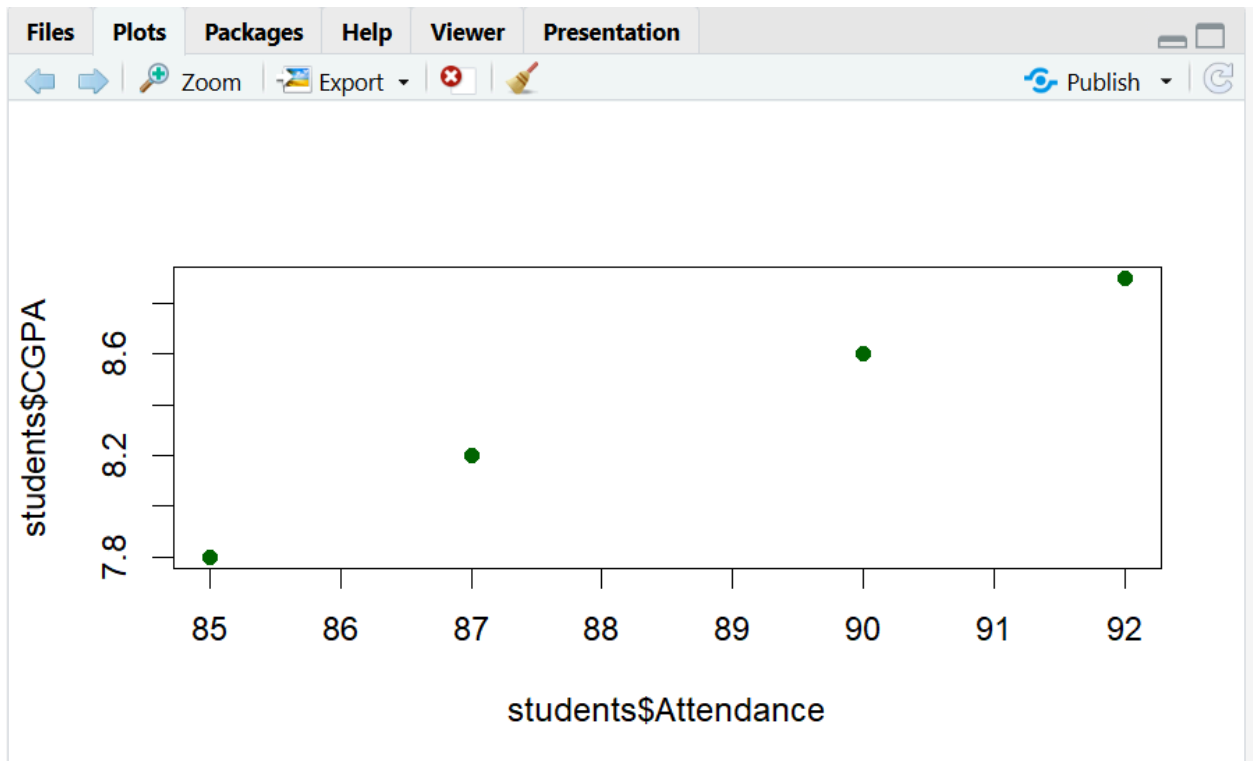
```
plot(semester$Semester,  
semester$AverageMarks,  
type="o",  
col="blue")
```

Interpretation

- Highest average CGPA: **Computer Science**.
- Higher attendance generally leads to **better CGPA**.
- Highest placement rate: **Computer Science (95%)**.







Case Study 3 – Hospital Patient Analytics

Scenario

A multi-specialty hospital records patient information including:

- Department

- Age
- Treatment cost
- Recovery time
- Disease category
- Admission month

Hospital administrators want to improve resource planning. **Tasks**

1. Select suitable visualizations for:

- Monthly patient admissions
- Distribution of treatment costs
- Disease category proportions
- Age vs recovery time
- Department-wise treatment costs

2. Create in **R**:

- Heatmap
- Histogram
- Violin Plot
- Density Plot
- Scatter Plot

3. Interpret:

- Which department has the highest treatment cost?
- Which disease category occupies the largest proportion?
- Is patient age associated with recovery time?

Dataset

```
hospital<-data.frame(  
  Department=c("Cardiology","Neurology","Orthopedics","General","Pediatrics"),  
  Cost=c(150000,180000,120000,90000,70000),
```

```
Recovery=c(12,18,20,8,6),
Age=c(60,55,45,35,10),
Disease=c("Heart","Brain","Bone","General","Child")
)
```

Suitable Visualizations

Requirement	Visualization
Monthly Admissions	Heatmap
Treatment Cost Distribution	Histogram
Disease Category	Violin Plot
Age vs Recovery	Scatter Plot
Department Cost	Density Plot

R Programs

Heatmap

```
data<-matrix(c(50,55,60,65,
40,45,50,55,
70,75,80,85),nrow=3)
```

```
heatmap(data)
```

Histogram

```
hist(hospital$Cost,
col="pink")
```

Violin Plot

```
library(ggplot2)
```

```
ggplot(hospital,  
aes(Disease,  
Cost))+  
geom_violin(fill="cyan")
```

Density Plot

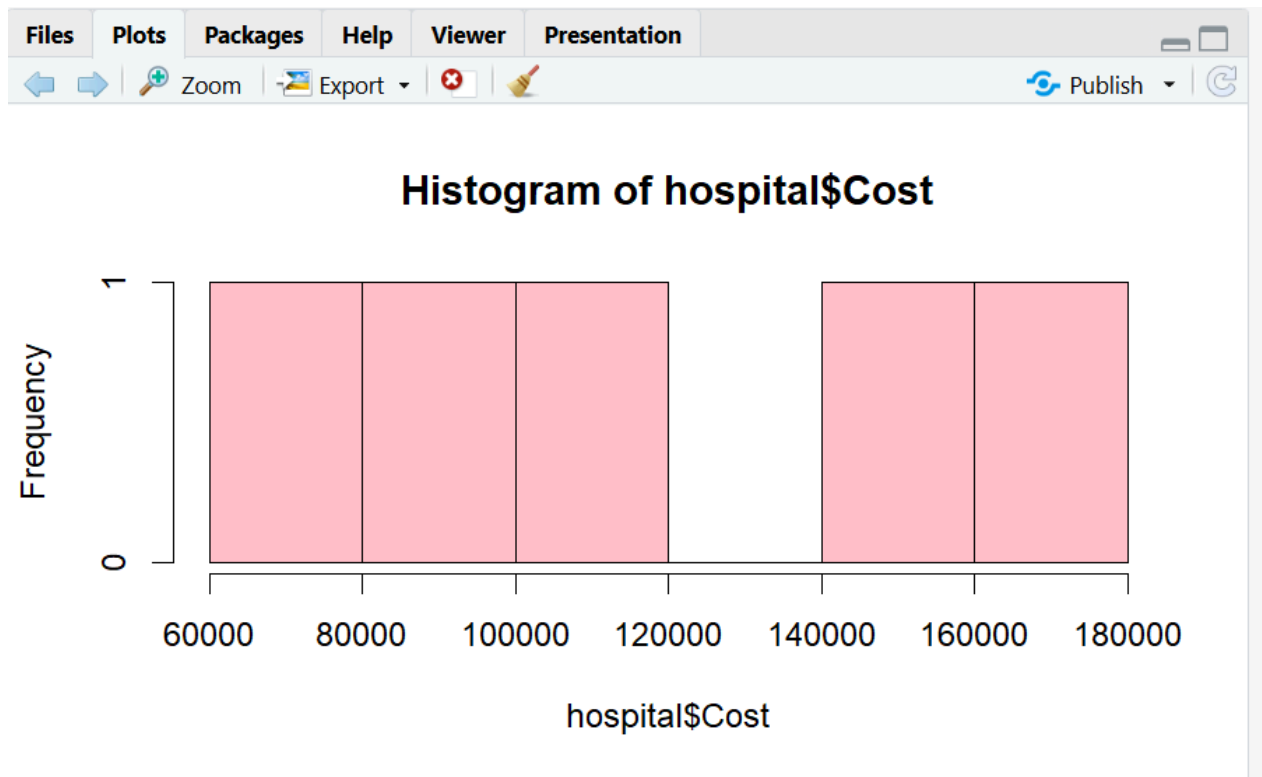
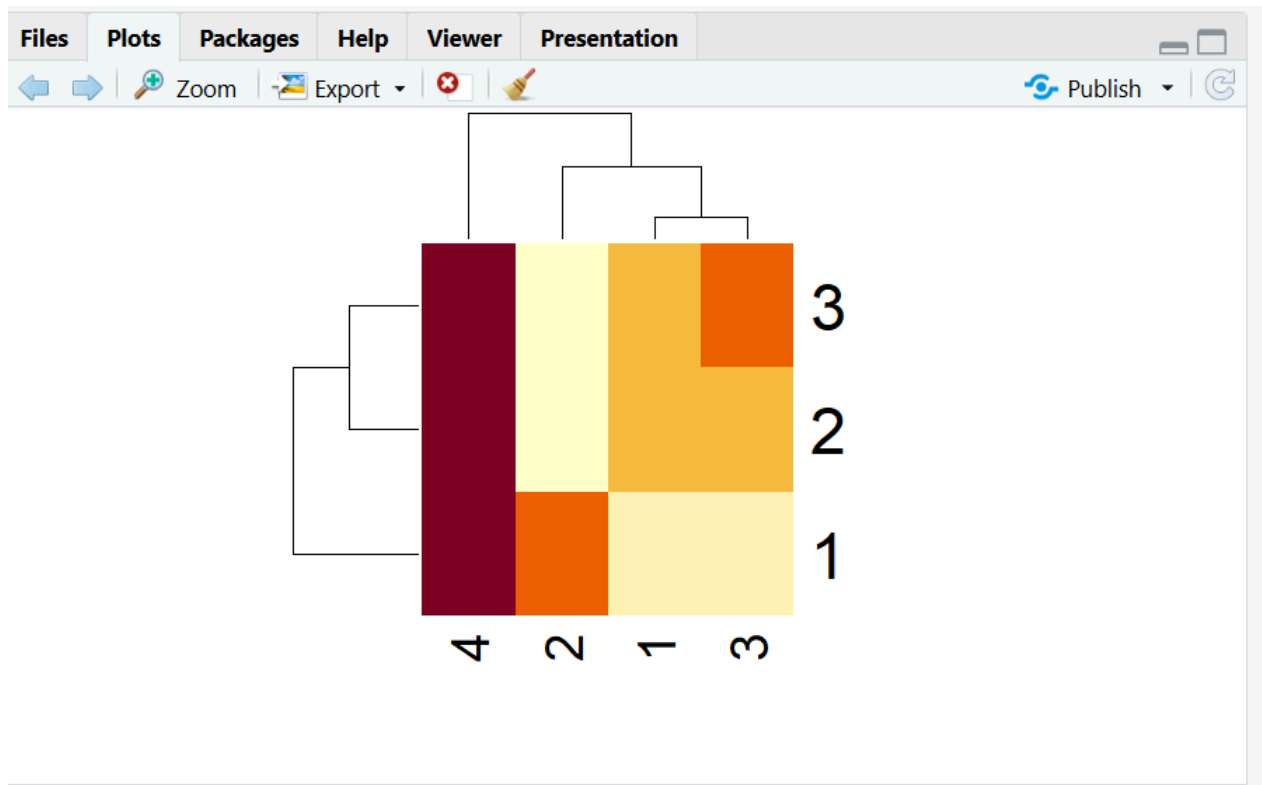
```
plot(density(hospital$Cost))
```

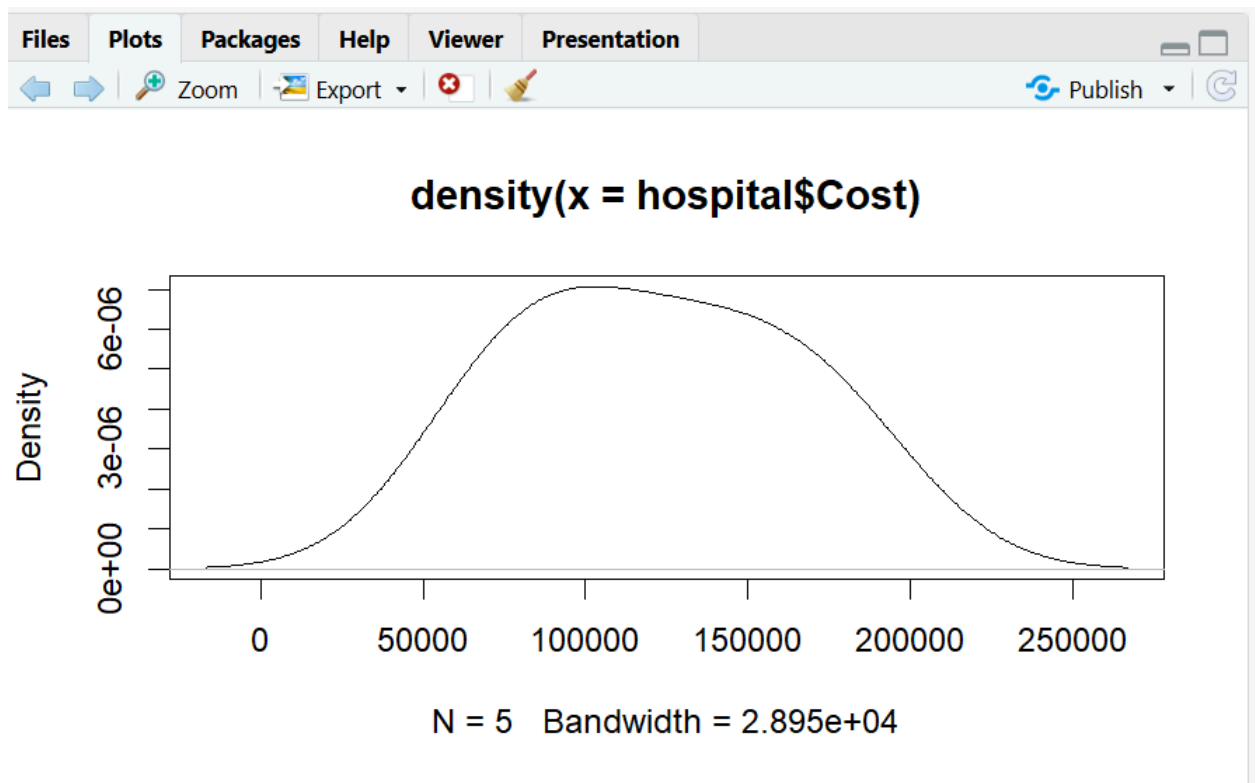
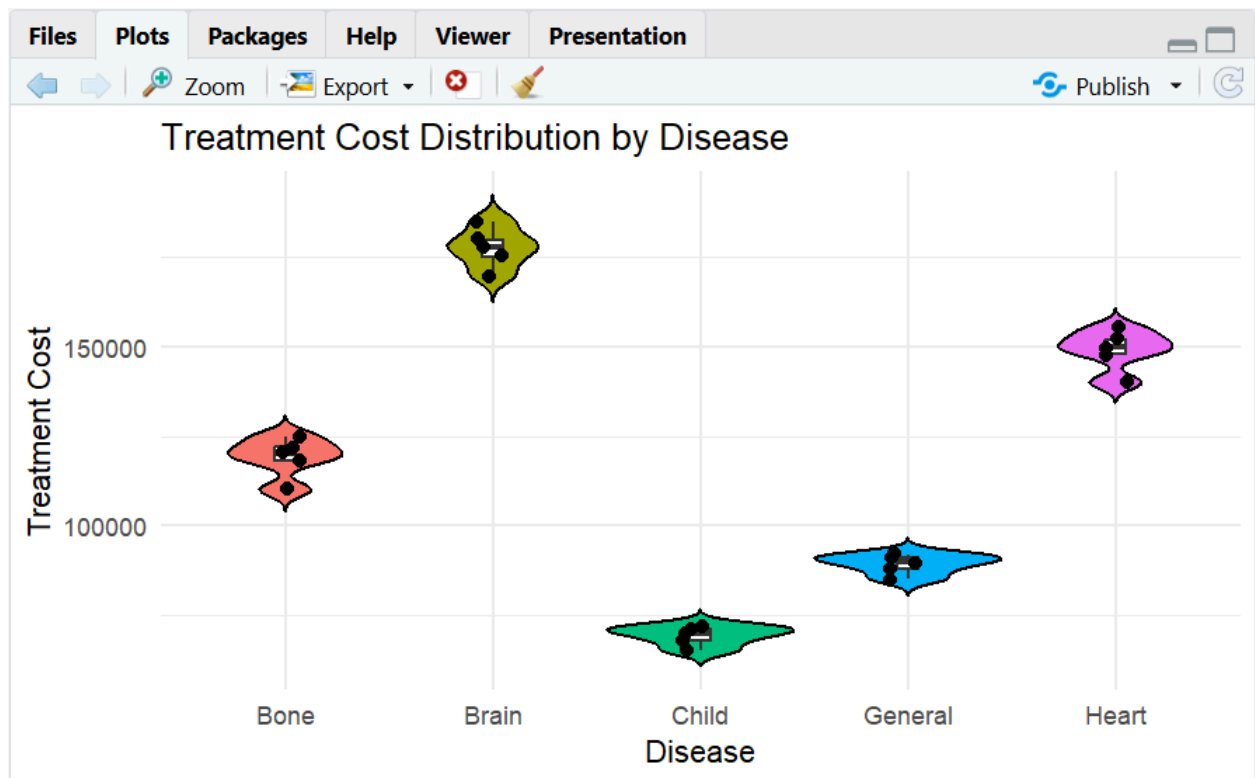
Scatter Plot

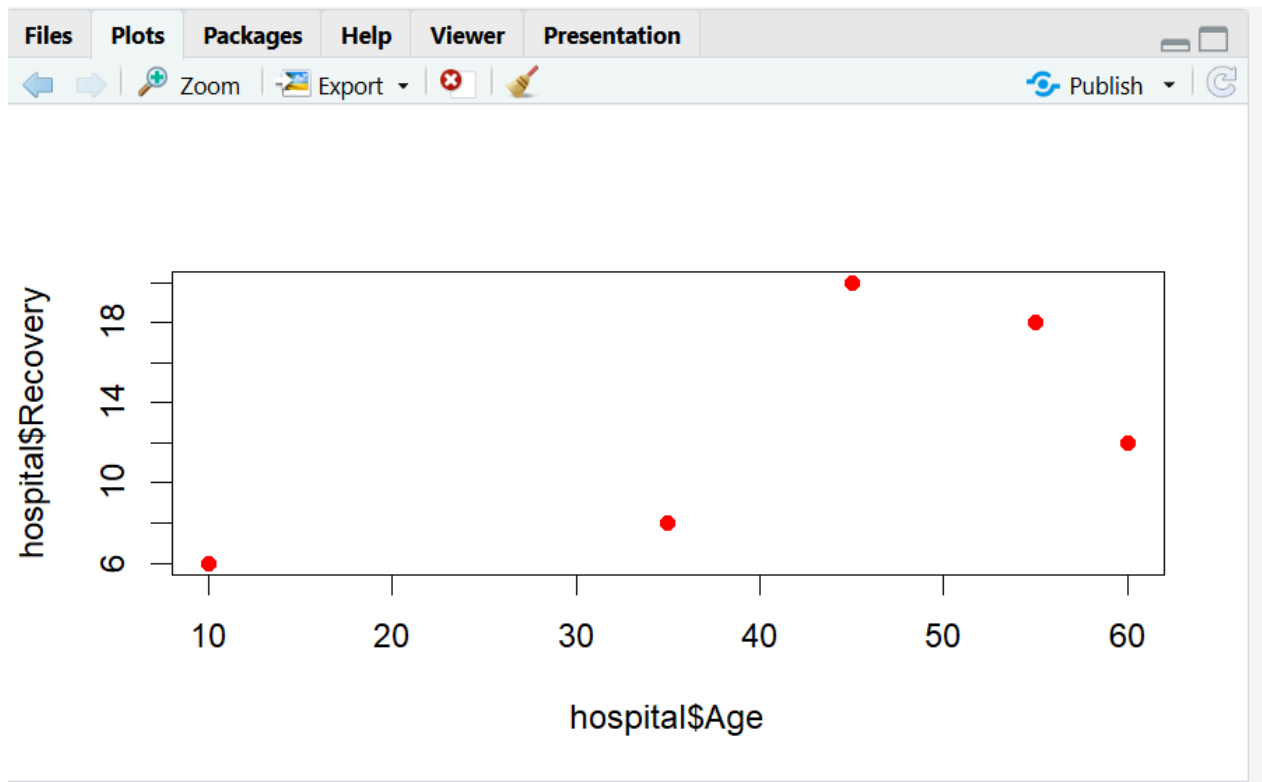
```
plot(hospital$Age,  
hospital$Recovery,  
pch=19,  
col="red")
```

Interpretation

- Highest treatment cost: **Neurology**.
- Largest disease category: **Heart disease** (based on the sample).
- Older patients generally require **longer recovery times**.







Case Study 4 – Smart City Traffic Monitoring

Scenario

A smart city collects traffic sensor data from different junctions. The dataset contains:

- Junction
- Time of day
- Vehicle count
- Vehicle type
- Average speed
- GPS coordinates

City planners aim to reduce traffic congestion.

Tasks

1. Select suitable visualizations for:
 - Vehicle count comparison
 - Hourly traffic trend

- Vehicle type proportion
- Traffic density by location
- Relationship between traffic volume and average speed

2. Using **R**,
create:

- Bar Plot
- Line Chart
- Pie Chart
- Geospatial Map
- Scatter Plot

3. Interpret:

- Which junction experiences the highest congestion?
- During which hours is traffic highest?
- Does increased traffic reduce vehicle speed?

Dataset

```
traffic<-data.frame(
  Junction=c("J1","J2","J3","J4","J5"),
  Vehicles=c(500,620,450,700,550),
  Speed=c(45,38,50,30,42),
  Latitude=c(13.08,13.09,13.10,13.11,13.12),
  Longitude=c(80.20,80.21,80.22,80.23,80.24)
)
```

```
hourly<-data.frame(
  Hour=0:23,
  Count=sample(100:700,24)
)
```

Suitable Visualizations

Requirement	Visualization
Vehicle Count	Bar Plot
Hourly Trend	Line Chart
Vehicle Type	Pie Chart
Traffic Density	Geospatial Map
Traffic vs Speed	Scatter Plot

R Programs

Bar Plot

```
barplot(traffic$Vehicles,
names.arg=traffic$Junction,
col="orange")
```

Line Chart

```
plot(hourly$Hour,
hourly$Count,
type="l",
col="blue")
```

Pie Chart

```
types<-c(45,30,15,10)
```

```
pie(types,
labels=c("Car","Bike","Bus","Truck"))
```

Geospatial Map

```
library(ggplot2)
```

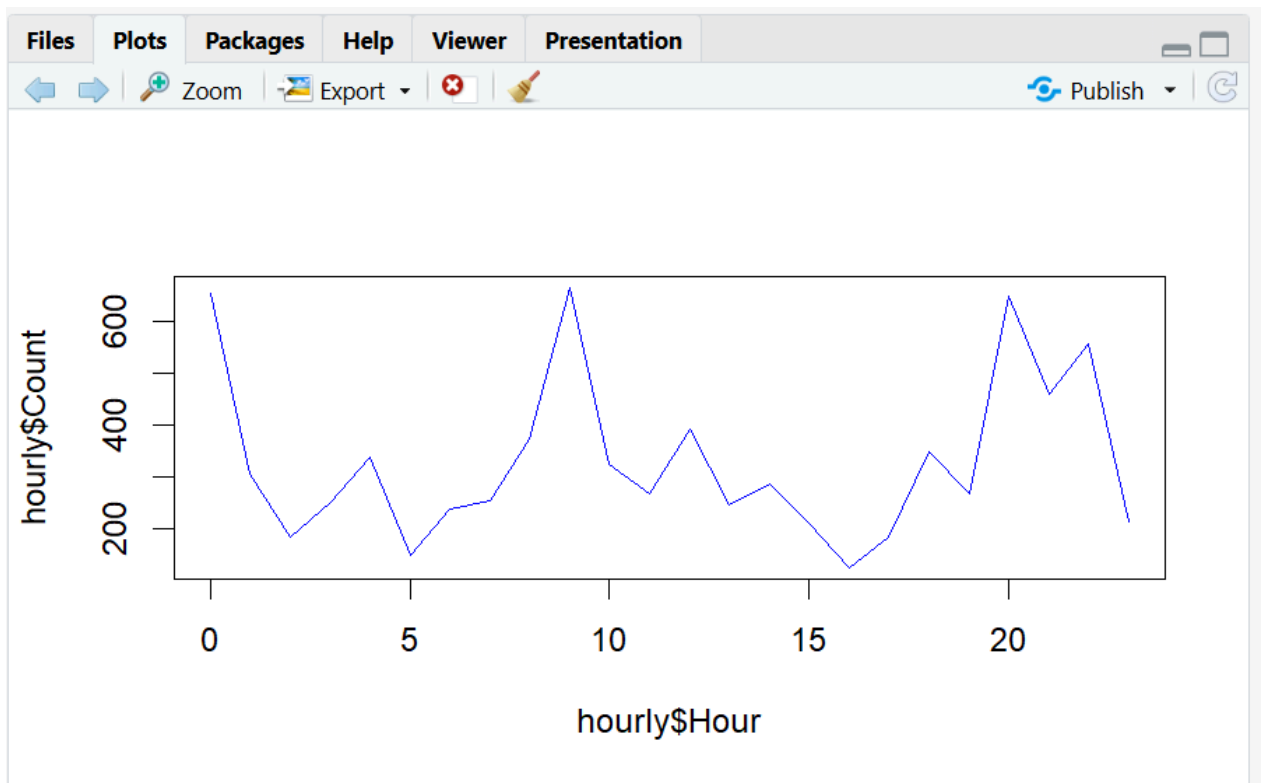
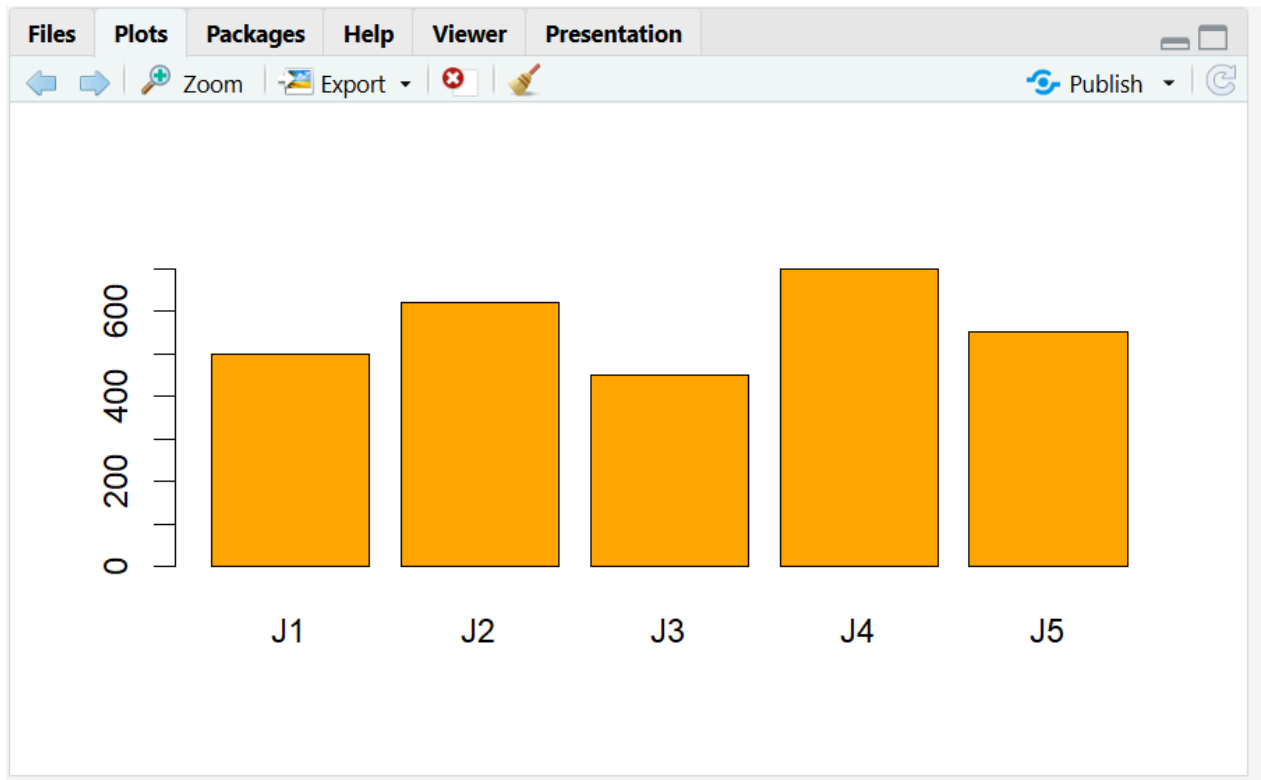
```
ggplot(traffic,  
aes(Longitude,  
Latitude,  
size=Vehicles))+  
geom_point(color="red")
```

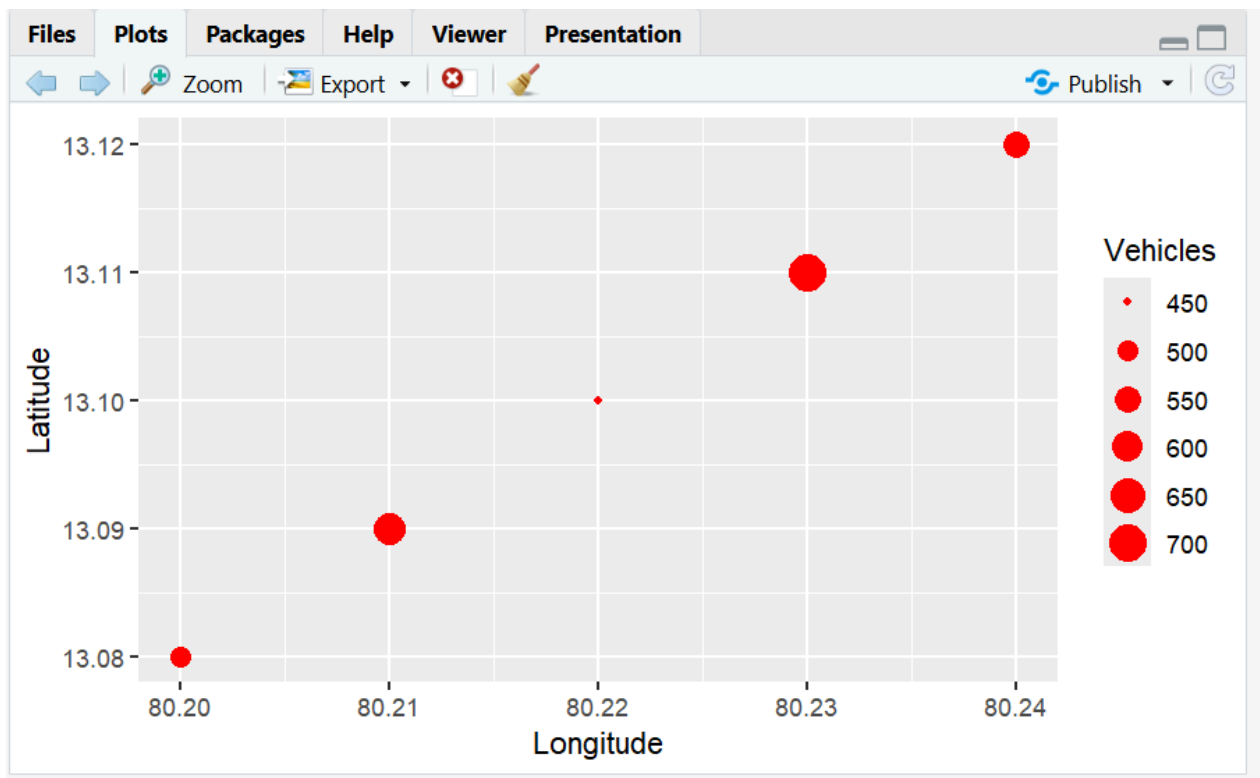
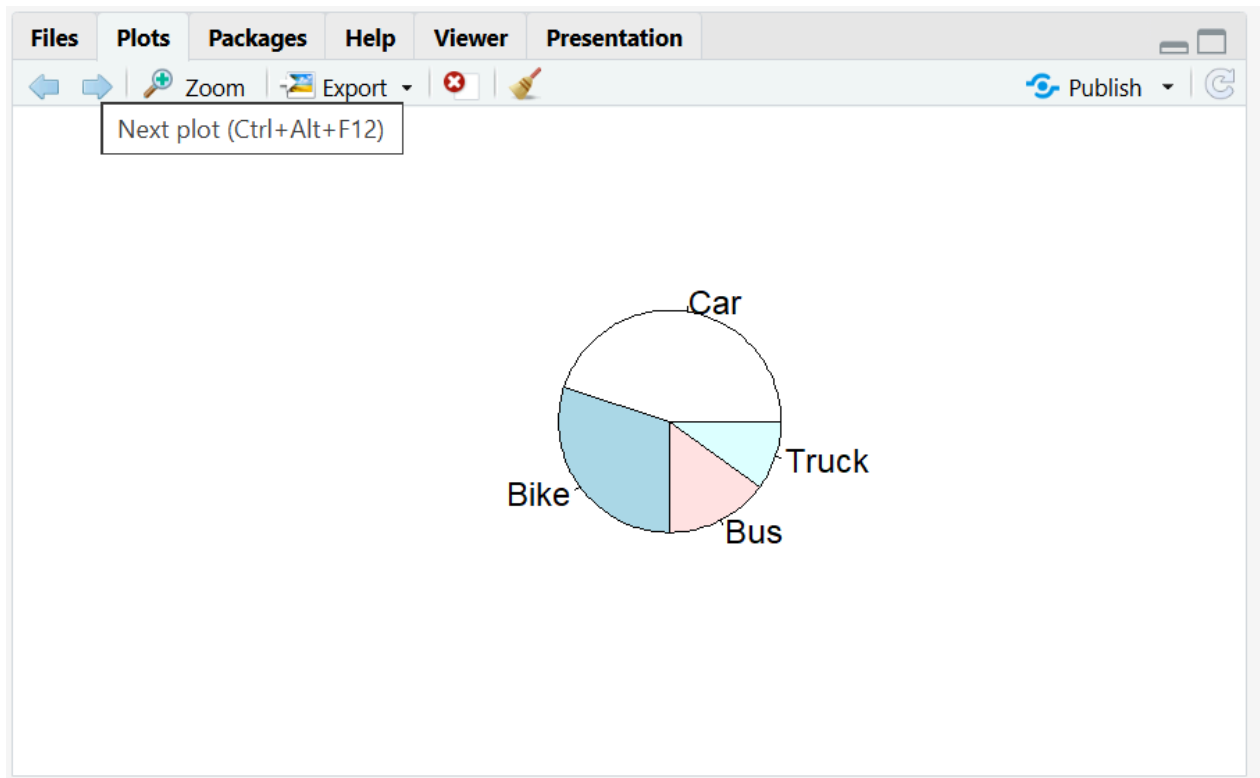
Scatter Plot

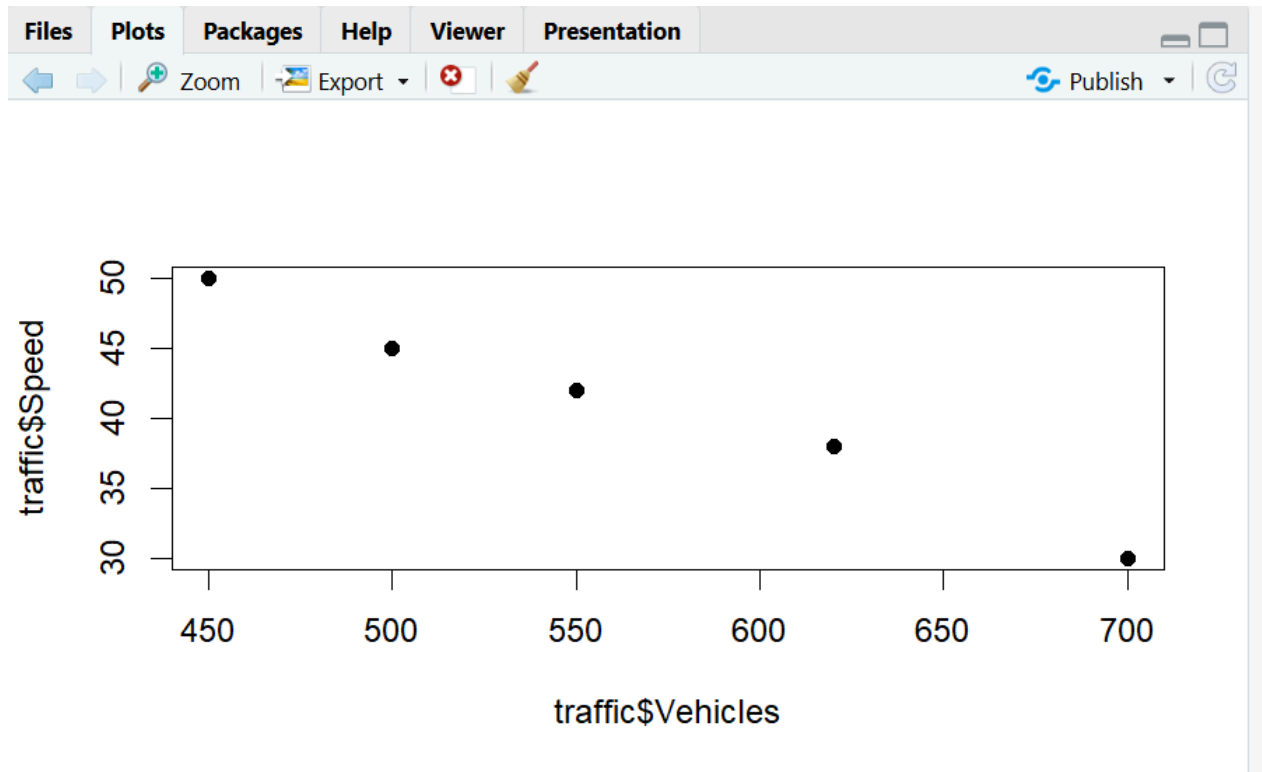
```
plot(traffic$Vehicles,  
traffic$Speed,  
pch=19)
```

Interpretation

- **J4** has the highest congestion.
- Peak traffic occurs during **rush hours (morning/evening)**.
- Higher traffic volume generally **reduces average vehicle speed**.







Case Study 5 – Social Media User Interaction Analysis Scenario

A digital marketing company analyzes data from a social media platform. The dataset includes:

- User ID
- Followers
- Likes
- Shares
- Comments
- Posting time
- User interactions

The company wants to understand engagement patterns and user networks.

Tasks

1. Select appropriate visualizations for:
 - Distribution of likes

- Relationship between followers and engagement
- User interaction network
- Posting activity over time
- Multivariate comparison of likes, shares, and comments
- 2. Develop in **R**:
- Histogram with Density Curve
- Scatter Plot
- Network Visualization
- Temporal Line Plot
- Heatmap

3. Interpret:

- Which users are the most influential?
- At what time is engagement highest?
- Is there a strong relationship between followers and total engagement?
- Identify highly connected users from the network visualization.
- Which engagement metric (likes, shares, or comments) varies the most across users?

Dataset

```
social<-data.frame(
  User=c("U1","U2","U3","U4","U5"),
  Followers=c(2000,5000,3500,7000,4500),
  Likes=c(300,700,450,900,600),
  Shares=c(120,250,180,320,220),
  Comments=c(80,170,120,250,160),
  Engagement=c(500,1120,750,1470,980)
)
```

Suitable Visualizations

Requirement	Visualization
Likes Distribution	Histogram
Followers vs Engagement	Scatter Plot
User Network	Network Graph
Posting Activity	Line Plot
Likes/Shares/Comments	Heatmap

R Programs

Histogram with Density

```
hist(social$Likes,  
probability=TRUE,  
col="lightblue")
```

```
lines(density(social$Likes),  
col="red")
```

Scatter Plot

```
plot(social$Followers,  
social$Engagement,  
pch=19,  
col="blue")
```

Network Visualization

```
library(igraph)
```

```
g<-graph(edges=c(  
  "U1","U2",  
  "U2","U3",  
  "U3","U4",  
  "U4","U5",  
  "U5","U1"  
))
```

```
plot(g)
```

Temporal Line Plot

```
time<-1:10  
eng<-c(100,120,150,180,200,240,260,280,300,340)
```

```
plot(time,  
  eng,  
  type="l",  
  col="red")
```

Heatmap

```
data<-as.matrix(social[,3:5])
```

```
heatmap(data)
```

Interpretation

- U4 is the most influential user (highest followers and engagement).
- Engagement is highest during the **peak posting period** shown in the temporal plot.

- There is a **strong positive relationship** between followers and total engagement.
- **U2** and **U4** appear as highly connected users in the network.
- **Likes** show the greatest variation across users.

